

# OneAgent and ActiveGate Migration

Series: M2S | Notebook: 6 of 8 | Created: January 2026

## Table of Contents

- 1. [Introduction](#)
- 2. [ActiveGate Migration](#)
- 3. [OneAgent Migration Strategies](#)
- 4. [Migration Execution](#)
- 5. [Validation](#)
- 6. [Next Steps](#)

## Prerequisites

Before starting this notebook, you should have:

| Requirement                | Description                        |
|----------------------------|------------------------------------|
| Completed M2S-01 to M2S-05 | Configurations migrated            |
| SaaS tenant ready          | Tenant provisioned and accessible  |
| Network connectivity       | Hosts can reach SaaS or ActiveGate |
| Deployment tokens          | SaaS installer tokens created      |
| Maintenance window         | Scheduled for migration            |

# Learning Objectives

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By the end of this notebook, you will:

- Deploy Environment ActiveGates for SaaS
  - Understand OneAgent migration strategies
  - Execute OneAgent reconfiguration
  - Validate successful agent migration
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## 1. Introduction

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Agent migration is the core technical step. OneAgents must be reconfigured to communicate with SaaS instead of Managed.

### Migration Options

| Option      | Method                          | Best For                  |
|-------------|---------------------------------|---------------------------|
| Reconfigure | Update existing OneAgent config | Most environments         |
| Reinstall   | Fresh OneAgent installation     | Major version upgrade     |
| Parallel    | Install new, remove old         | Zero-downtime requirement |

### Migration Sequence

1. Deploy Environment ActiveGate (if needed)
  2. Migrate non-production hosts first
  3. Validate data flow
  4. Migrate production hosts
  5. Final validation
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## 2. ActiveGate Migration

### 2.1 When ActiveGate is Required

| Scenario                         | ActiveGate Needed |
|----------------------------------|-------------------|
| Hosts can reach SaaS directly    | No (optional)     |
| Hosts behind firewall            | Yes, for routing  |
| Extensions 2.0 required          | Yes               |
| Private synthetic monitoring     | Yes               |
| VMware/cloud platform monitoring | Yes               |

### 2.2 Deploy Environment ActiveGate

#### Step 1: Download Installer

From Dynatrace SaaS UI:

1. Go to **Deploy Dynatrace → ActiveGate**
2. Select **Linux** or **Windows**
3. Copy the installer URL and token

#### Step 2: Linux Installation

```
# Download installer
wget -O Dynatrace-ActiveGate-Linux.sh
"https://{tenant}.live.dynatrace.com/api/v1/deployment/installer/gateway/unix
/latest?Api-Token={token}&arch=x86&flavor=default"

# Run installer
sudo /bin/sh Dynatrace-ActiveGate-Linux.sh
```

### Step 3: Windows Installation

```
# Download installer
Invoke-WebRequest -Uri
"https://{tenant}.live.dynatrace.com/api/v1/deployment/installer/gateway/wind
ows/latest?Api-Token={token}" -OutFile Dynatrace-ActiveGate.exe

# Run installer
.\Dynatrace-ActiveGate.exe
```

## 2.3 Verify ActiveGate Connection

```
// ActiveGate validation - use the Entities API v2 or Settings API
// Note: ActiveGates are not directly queryable via DQL fetch
// Use: GET /api/v2/entities?entitySelector=type("ENVIRONMENT_ACTIVE_GATE")
// Or check the ActiveGates page in the Dynatrace UI
```

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## 3. OneAgent Migration Strategies

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### 3.1 Strategy 1: Reconfigure (Recommended)

Update the existing OneAgent to point to SaaS.

**Pros:**

- No reinstallation required
- Minimal disruption
- Configuration preserved

**Cons:**

- Requires restart of OneAgent service
- Brief monitoring gap during restart

## 3.2 Strategy 2: Reinstall

Uninstall Managed OneAgent, install SaaS OneAgent.

**Pros:**

- Clean installation
- Opportunity to upgrade version

**Cons:**

- Longer monitoring gap
- More complex process

## 3.3 Strategy 3: Parallel Installation

Install SaaS OneAgent alongside Managed (temporarily).

**Pros:**

- Zero monitoring gap
- Easy rollback

**Cons:**

- Resource overhead during parallel period
- More complex coordination

## 3.4 Strategy Selection Matrix

| Requirement              | Recommended Strategy |
|--------------------------|----------------------|
| Minimal downtime         | Reconfigure          |
| Major version upgrade    | Reinstall            |
| Zero monitoring gap      | Parallel             |
| Risk-averse organization | Parallel             |
| Limited change windows   | Reconfigure          |

## 4. Migration Execution

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### 4.1 Reconfigure OneAgent (Linux)

#### Option A: Using oneagentctl

```
# Get current configuration
sudo /opt/dynatrace/oneagent/agent/tools/oneagentctl --get-server

# Set new SaaS server
sudo /opt/dynatrace/oneagent/agent/tools/oneagentctl --set-
server="https://{tenant}.live.dynatrace.com:443/communication"

# Set tenant token (get from SaaS UI)
sudo /opt/dynatrace/oneagent/agent/tools/oneagentctl --set-tenant-token="
{tenant-token}"

# Restart OneAgent
sudo systemctl restart oneagent
```

#### Option B: Using Configuration File

```
# Edit configuration
sudo vi /var/lib/dynatrace/oneagent/agent/config/hostcustomproperties.conf

# Add/update:
# Server=https://{tenant}.live.dynatrace.com:443/communication
# TenantToken={tenant-token}

# Restart
sudo systemctl restart oneagent
```

## 4.2 Reconfigure OneAgent (Windows)

```
# Get current configuration
& "C:\Program Files\dynatrace\oneagent\agent\tools\oneagentctl.exe" --get-server

# Set new SaaS server
& "C:\Program Files\dynatrace\oneagent\agent\tools\oneagentctl.exe" --set-server="https://{tenant}.live.dynatrace.com:443/communication"

# Set tenant token
& "C:\Program Files\dynatrace\oneagent\agent\tools\oneagentctl.exe" --set-tenant-token="{tenant-token}"

# Restart service
Restart-Service Dynatrace*
```

## 4.3 Via ActiveGate Routing

If using ActiveGate for routing:

```
# Set ActiveGate as communication endpoint
sudo /opt/dynatrace/oneagent/agent/tools/oneagentctl --set-server="https://{activegate-ip}:9999/communication"

# Set tenant token
sudo /opt/dynatrace/oneagent/agent/tools/oneagentctl --set-tenant-token="{tenant-token}"

# Restart
sudo systemctl restart oneagent
```

## 4.4 Critical: Application Process Restarts

 **IMPORTANT:** After migrating OneAgent, application processes need to be restarted for full monitoring capabilities.

**When Restarts Are Required:**

| OneAgent Mode                | Restart Required?                | What Happens Without Restart                                            |
|------------------------------|----------------------------------|-------------------------------------------------------------------------|
| <b>Full-stack monitoring</b> | ✅ <b>Yes</b>                     | Services won't appear; no distributed tracing                           |
| <b>Infrastructure-only</b>   | ✅ <b>Yes</b> (for some features) | Application Security vulnerability detection and JMX metrics won't work |

### Full-Stack Monitoring (Process Restarts Required):

| Technology | Why Restart Required                         |
|------------|----------------------------------------------|
| Java       | Code modules inject at JVM startup           |
| .NET       | Profiler attaches at process start           |
| Node.js    | Instrumentation loads at startup             |
| PHP        | Zend extension loads at Apache/PHP-FPM start |
| Go         | Instrumentation compiles at build time       |

### What happens without restart:

- Host-level metrics will flow (CPU, memory, disk)
- **Application-level monitoring will NOT work**
- No distributed tracing
- No code-level visibility
- No service detection for existing processes

### Infrastructure-Only Mode (Restarts Recommended):

Even in infrastructure-only mode, restarts enable:

- Application Security vulnerability detection
- JMX metrics collection
- Some process-level insights

### Recommended Approach:

1. Migrate OneAgent during maintenance window
2. Perform rolling restart of application services
3. Or schedule next regular deployment to restart processes



```
# Example: Restart Java application
sudo systemctl restart myapp.service

# Example: Restart Apache/PHP
sudo systemctl restart apache2

# Example: Restart Node.js application
pm2 restart all
```

## 4.5 Bulk Migration with Automation

For large environments, use configuration management:

### Ansible Example:

```
- name: Reconfigure OneAgent for SaaS
  hosts: all_monitored_hosts
  become: yes
  tasks:
    - name: Set SaaS server
      command: /opt/dynatrace/oneagent/agent/tools/oneagentctl --set-
server="https://{{ saas_tenant }}.live.dynatrace.com:443/communication"

    - name: Set tenant token
      command: /opt/dynatrace/oneagent/agent/tools/oneagentctl --set-tenant-
token="{{ tenant_token }}"

    - name: Restart OneAgent
      systemd:
        name: oneagent
        state: restarted

    - name: Restart application services (for full-stack monitoring)
      systemd:
        name: "{{ item }}"
        state: restarted
      loop: "{{ application_services }}"
      when: restart_apps | default(false)
```

## 4.6 OneAgent Re-homing Considerations

When using the `oneagentctl` reconfigure approach:

| Preserved               | Not Preserved                        |
|-------------------------|--------------------------------------|
| Host ID                 | N/A (stays same)                     |
| Process group IDs       | N/A (regenerated based on detection) |
| Service IDs             | N/A (regenerated based on detection) |
| Custom metadata         | ✅ Preserved                          |
| Environment tags        | ✅ Preserved                          |
| Host group assignment   | ✅ Preserved                          |
| Network zone assignment | Must reconfigure if changed          |

**Tip:** Using `oneagentctl` to reconfigure (rather than reinstall) preserves host identity and custom metadata.

## 5. Validation

### 5.1 Immediate Validation

After migrating hosts, verify they appear in SaaS:

```
// Count hosts reporting to SaaS
fetch dt.entity.host
| summarize totalHosts = count()
```

```
// List hosts reporting to SaaS
fetch dt.entity.host
| fieldsAdd hostName = entity.name
| fields hostName, id
| sort hostName desc
| limit 20
```

### 5.1.1 OneAgent Version Check

**Note:** OneAgent version is not available as a DQL field. Use the Dynatrace UI: Navigate to **Manage** → **Deployment status** → **OneAgent** to view version distribution.

## 5.2 Data Flow Validation

Verify metrics are flowing:

```
// Check for recent CPU metrics
timeseries avg(dt.host.cpu.usage), by:{dt.entity.host}
| limit 10
```

```
// Verify logs are flowing
fetch logs
| summarize logCount = count(), by:{time_bucket = bin(timestamp, 5m)}
| sort time_bucket desc
| limit 12
```

## 5.3 Service Discovery Validation

```
// Check active services discovered after migration
fetch dt.entity.service
| fieldsAdd serviceName = entity.name
| summarize activeServices = count()
```

## 5.4 Migration Completion Checklist

| Validation             | Query/Method                   | Status |
|------------------------|--------------------------------|--------|
| All hosts reporting    | Entity count matches expected  | [ ]    |
| CPU metrics flowing    | Timeseries query returns data  | [ ]    |
| Memory metrics flowing | Timeseries query returns data  | [ ]    |
| Logs ingesting         | Log count > 0                  | [ ]    |
| Services discovered    | Service count matches expected | [ ]    |
| No OneAgent errors     | Check OneAgent logs            | [ ]    |

## Troubleshooting

### Common Issues

| Issue                | Symptom                | Solution                  |
|----------------------|------------------------|---------------------------|
| Connection refused   | Host offline in SaaS   | Check firewall rules      |
| Invalid token        | OneAgent errors in log | Regenerate tenant token   |
| DNS resolution       | Can't reach endpoint   | Verify DNS settings       |
| Certificate errors   | SSL/TLS failures       | Check proxy configuration |
| ActiveGate not found | Routing failures       | Verify AG is running      |

### OneAgent Log Locations

| Platform | Log Path                               |
|----------|----------------------------------------|
| Linux    | /var/log/dynatrace/oneagent/           |
| Windows  | C:\ProgramData\dynatrace\oneagent\log\ |

## Rollback Procedure

If migration fails, revert to Managed:

```
# Revert to Managed server
sudo /opt/dynatrace/oneagent/agent/tools/oneagentctl --set-
server="https://{managed-cluster}/communication"
sudo /opt/dynatrace/oneagent/agent/tools/oneagentctl --set-tenant-token="
{managed-token}"
sudo systemctl restart oneagent
```

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## 6. Next Steps

### Immediate Actions

1. **Deploy ActiveGate** - If required for your architecture
2. **Migrate test hosts** - Start with non-production
3. **Validate thoroughly** - Run all validation queries
4. **Migrate production** - After successful test
5. **Document issues** - Track any problems encountered

### Continue the Series

| Next Notebook              | Focus                            |
|----------------------------|----------------------------------|
| M2S-07: Security & Privacy | Security considerations for SaaS |

### Agent Resources

- [OneAgent Documentation](#)
  - [ActiveGate Documentation](#)
  - [OneAgentCtl Reference](#)
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# Summary

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In this notebook, you learned:

- How to deploy Environment ActiveGates for SaaS
- OneAgent migration strategies and when to use each
- Step-by-step reconfiguration procedures
- Validation queries to confirm successful migration
- Troubleshooting common migration issues

**Key Takeaway:** The reconfigure strategy is simplest for most migrations. Use `oneagentctl` to update the server endpoint and tenant token, then restart the service.

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*Continue to M2S-07: Security & Privacy for security considerations in SaaS.*

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